HW #1

1. a) $x(n) = \sin\left(\frac{\pi n}{4} + \frac{\pi}{8}\right)$

$N = \frac{2k\pi}{w} = \frac{2k\pi}{\frac{\pi}{4}} = 8k$ fundamental period $N = 8$, **periodic**

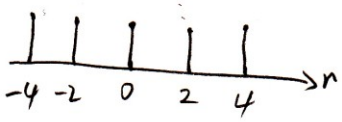
b). $x(n) = \sin\left(\frac{\pi n}{4} + \frac{\pi}{8}\right) + \cos\left(\frac{\pi}{3}n\right)$

$N_1 = \frac{2k\pi}{w_1} = \frac{2k\pi}{\frac{\pi}{4}} = 8k \quad N_2 = \frac{2k\pi}{\frac{\pi}{3}} = 6k$

$\frac{N_1}{N_2} = \frac{8k}{6k} = \frac{4}{3} \Rightarrow N = 3N_1 = 3\times 8 = 24$ **periodic**

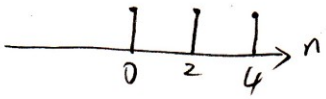
c). $x(n) = \sum_{m=-\infty}^{\infty} \delta(n-2m)$

$\delta(n-2m) = \begin{cases} 1 & n = 2m \\ 0 & n \neq 2m \end{cases}$



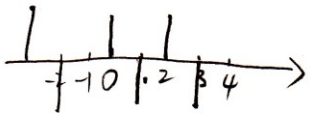
$\Rightarrow N = 2$ **periodic**

d). $x(n) = \sum_{m=0}^{\infty} \delta(n-2m)$



nonperiodic

e). $x(n) = \sum_{m=-\infty}^{\infty} (-1)^m \delta(n-m)$



periodic $N = 2$

$\sum_{n=N_1}^{\infty} q^n = \frac{q^{N_1}}{1-q} \quad |q| < 1$

2. a). $x(n) = 2\left(\frac{1}{3}\right)^n \quad n > 0$

$E = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \sum_{n=0}^{\infty} \left|2\left(\frac{1}{3}\right)^n\right|^2 = \sum_{n=1}^{\infty} 4\cdot\left(\frac{1}{9}\right)^n = 4\sum_{n=1}^{\infty}\left(\frac{1}{9}\right)^n = 4\cdot\frac{\left(\frac{1}{9}\right)^1}{1-\frac{1}{9}} = \frac{1}{2}$

So this signal is an energy signal

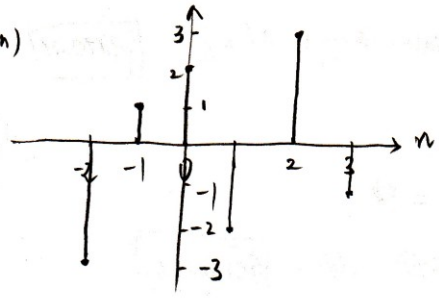
b). $x(n) = u(n)$

$P = \lim_{N\to\infty} \frac{1}{N+1}\sum_{n=0}^{N} |u(n)|^2 = \lim_{N\to\infty} \frac{1}{N+1}\sum_{n=0}^{N} 1 = 1$

So this signal is a power signal

3. $x(n) = [-3, 1, \underset{\uparrow}{2}, -2, 3, -1]$

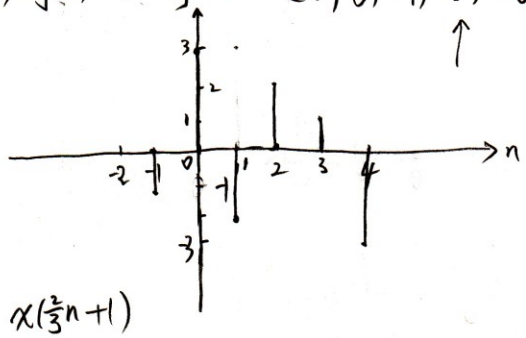
a). $x(n)$



b). $x(2-n)$

$y(n) = x(-n)$, $y(n) = [-1, 3, -2, \underset{\uparrow}{2}, 1, -3]$

$z(n) = y(n-2) = x[-(n-2)]$ $z(n) = [0, 0, -1, \underset{\uparrow}{3}, -2, 2, 1, -3]$



c). $x(\frac{2}{3}n+1)$

$y(n) = x(\frac{2}{3}n+1)$

$y(-3) = x(-2+1) = x(-1) = 1$

$y(-2) = x(-\frac{4}{3}+1) = 0$

$y(-1) = x(-\frac{2}{3}+1) = 0$

$y(0) = x(0+1) = x(1) = -2$

$y(1) = x(\frac{2}{3}+1) = 0$

$y(2) = x(\frac{4}{3}+1) = 0$

$y(3) = x(2+1) = x(3) = -1$

